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PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 31-01-2026**

Course Name: Cybersphere and Security(CSS) **Page no:**

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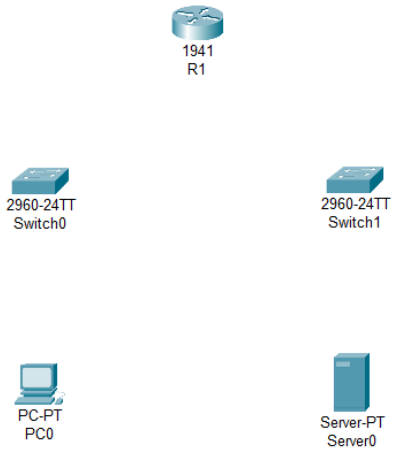
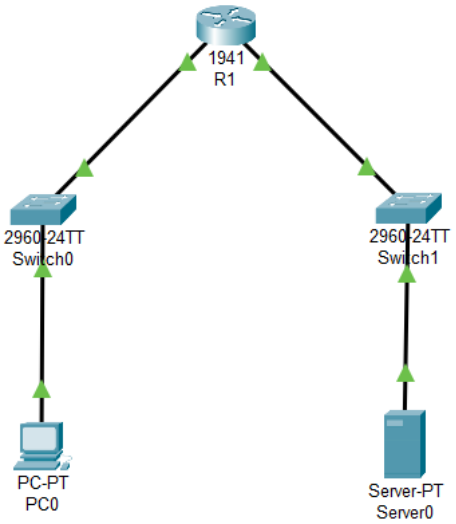
Q. Configure the IOS Intrusion Prevention System (IPS) Using the CLI.

Steps:

1. Open the Cisco packet
2. Add the one PC(PC0), two switches of 2960-24TT(Switch0, Switch1), one router of 1941(R1), and one server



3. Now connect copper straight-through wire to devices as given below.
   - **PC0 → Switch0:** PC0 – FastEthernet0 → Switch0 FastEthernet0/1
   - **Switch0 → Router(R1):** Switch0 FastEthernet0/2 → R1 GigabitEthernet0/0
   - **Router(R1) → Switch1:** R1 GigabitEthernet0/1 → Switch1 FastEthernet0/1
   - **Switch1 → Server0:** Switch1 FastEthernet0/2 → Server0 FastEthernet0



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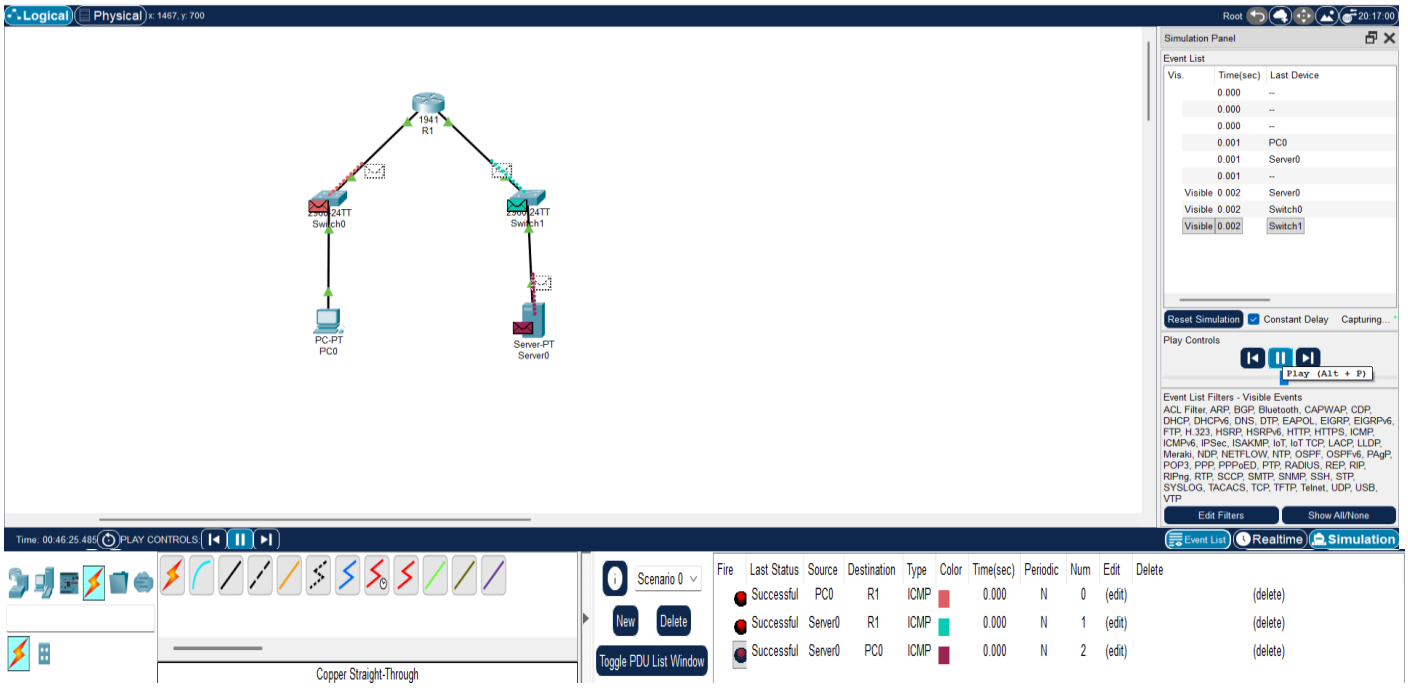
4. Assign the addresses as follow:
   - To assign addresses to Server0, PC0 go to the desktop → IP configuration.
   - For R1 go to config.

IP Config	IPv4 Address	Subnet Mask	Default Gateway
PC0	192.168.0.2	255.255.255.0	192.168.0.1
R1 G0/0	192.168.0.1	255.255.255.0	-
R1 G0/1	10.10.10.1	255.255.255.252	-
Server0	10.10.10.2	255.255.255.252	10.10.10.1

5. Now communication is allowed in the network.



6. Click on R1 → CLI → paste these lines in that.
   - show version

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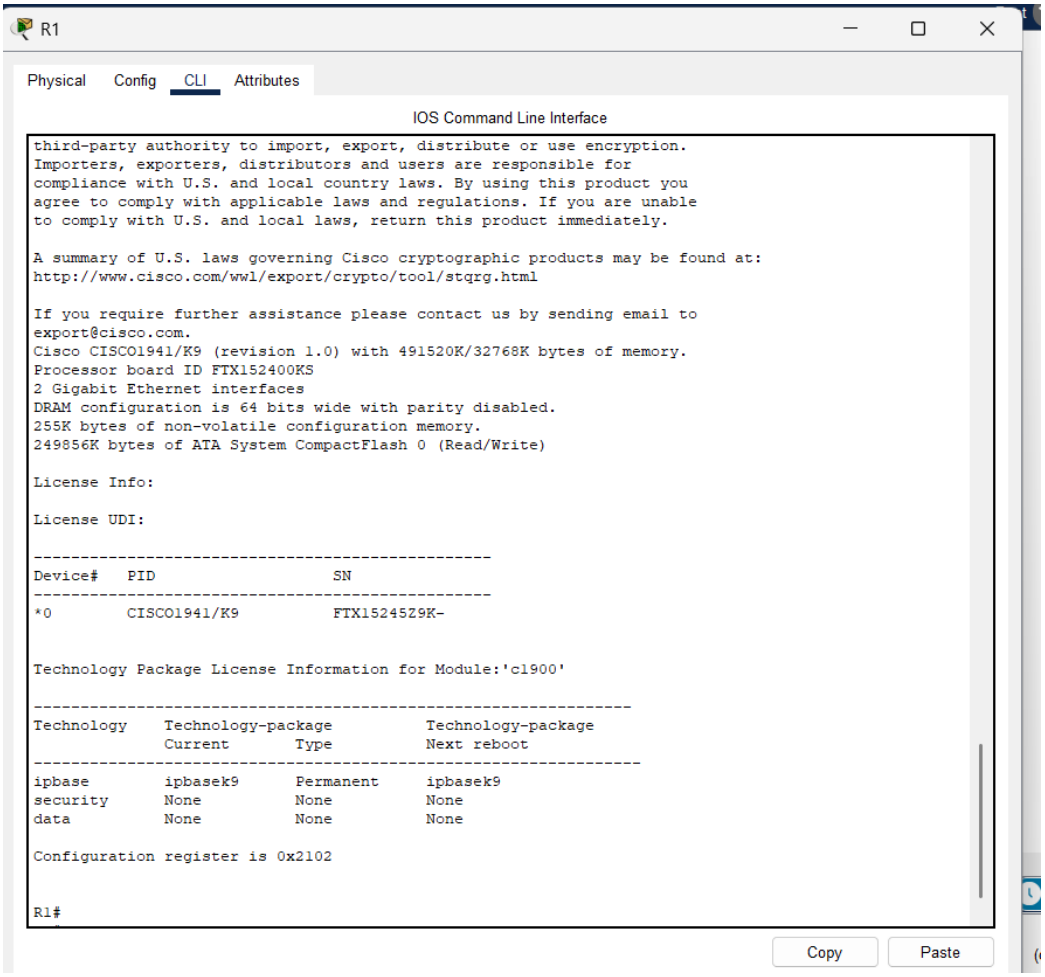

- license boot module c1900 technology-package securityk9

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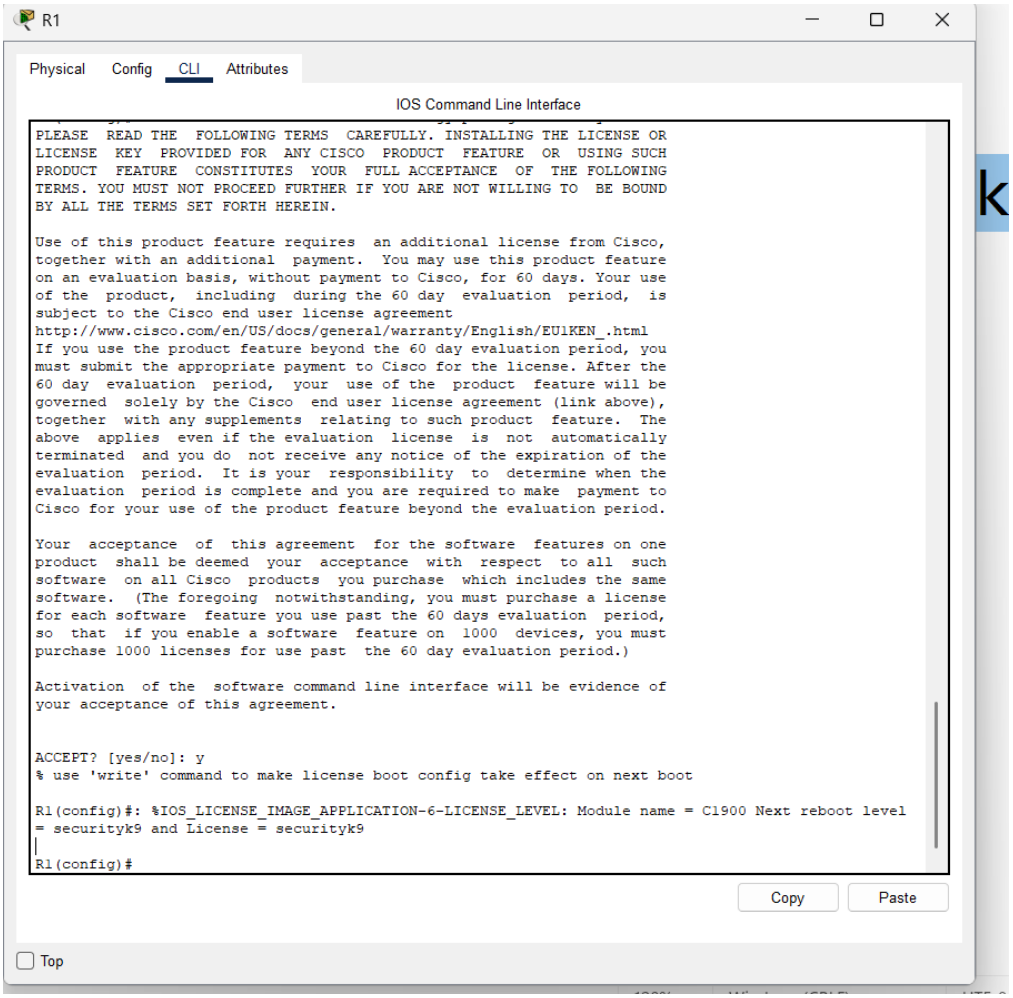

- ex
- reload
    - when asked for confirm hit enter.

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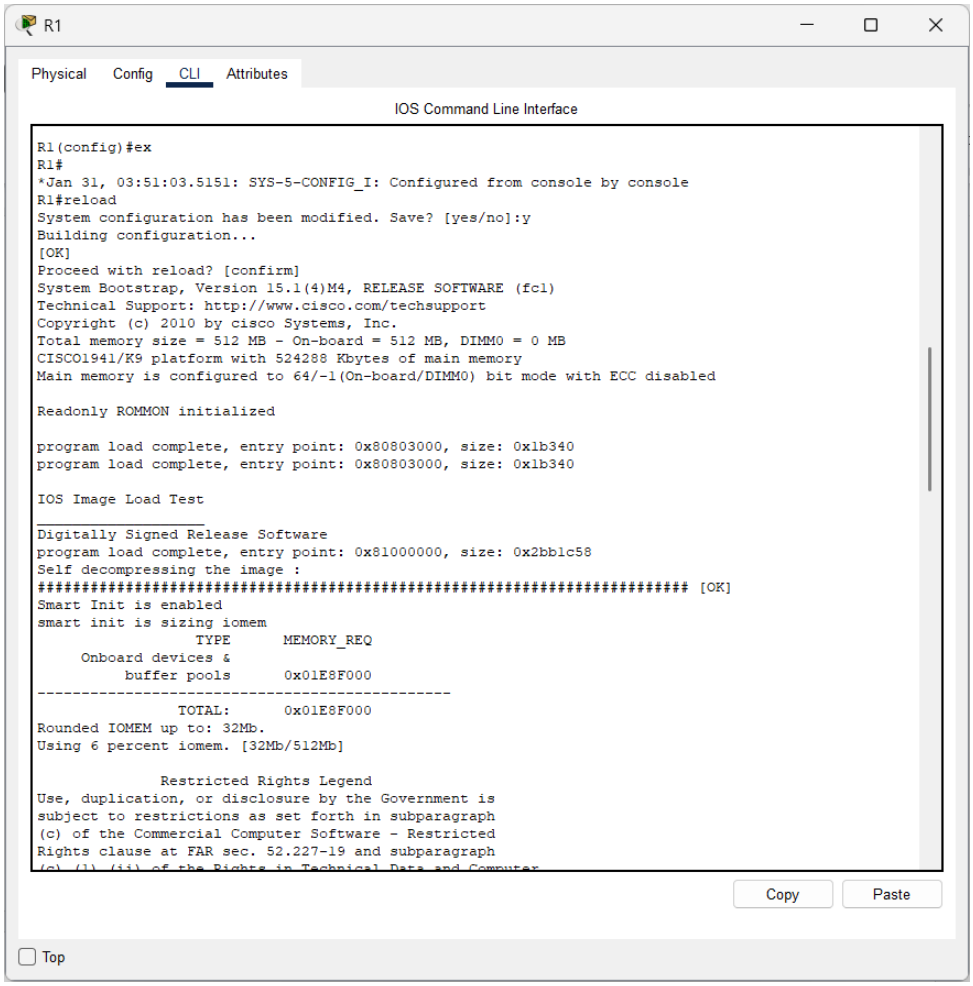

- en
- show version

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R1

Physical Config CLI Attributes

IOS Command Line Interface

```
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

License Info:

License UDI:

-------------------------------------------------
Device#   PID                   SN
-------------------------------------------------
*0        CISCO1941/K9          FTX15245Z9K-


Technology Package License Information for Module:'c1900'

----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      securityk9    Evaluation    securityk9
data          disable       None          None

Configuration register is 0x2102


R1#
```

- mkdir iosips

```
R1#
R1#mkdir iosips
Create directory filename [iosips]?
Created dir flash:iosips

R1#
```

- conf t
- ip ips config location flash:iosips
- ip ips name iosips
- ip ips signature-category
- category all
- retired true
- ?

Teacher's Signature

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip ips config location flash:iosips
R1(config)#ip ips name iosips
R1(config)#ip ips signature-category
R1(config-ips-category)#category all
R1(config-ips-category-action)#retired true
R1(config-ips-category-action)#ex
R1(config-ips-category)#
R1(config-ips-category)#?
  category  Category keyword
  exit      Exit from Category Mode
  no        Negate or set default values of a command
R1(config-ips-category)#
```

- ex
- category ios_ips basic
- retired false
- ex
- ex
    - When asked for confirm hit enter

```
R1(config-ips-category-action)#ex
R1(config-ips-category)#category ios_ips basic
R1(config-ips-category-action)#retired false
R1(config-ips-category-action)#ex
R1(config-ips-category)#ex
Do you want to accept these changes? [confirm]
Applying Category configuration to signatures ...
%IPS-6-ENGINE_BUILDING: atomic-ip - 288 signatures - 6 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 30 ms - packets for this engine will be scanned

R1(config)#
```

- int gi 0/1
- ip ips iosips out
- ex
- ip ips notify log
- ex

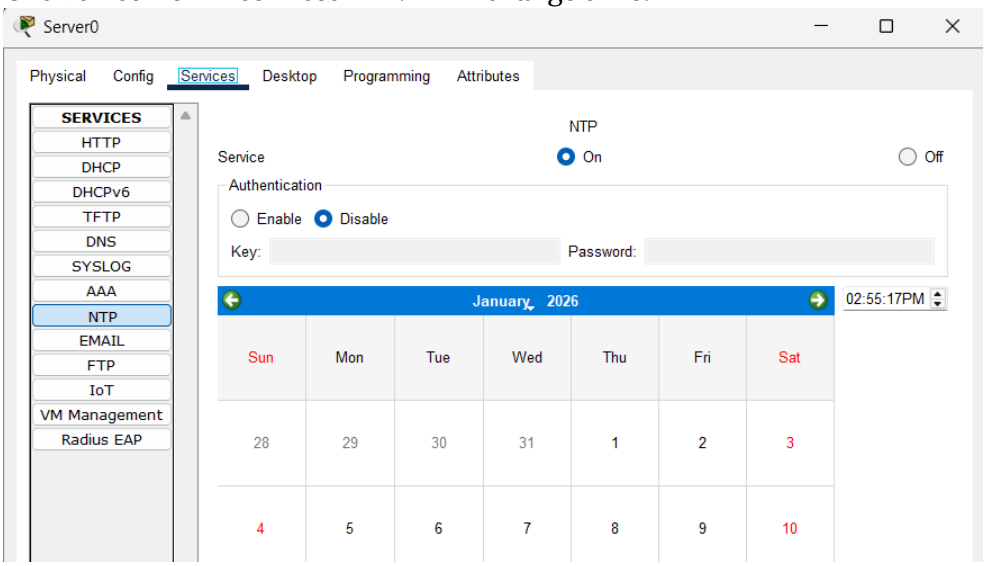
R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1(config)#int gi 0/1
R1(config-if)#ip ips iosips out
R1(config-if)#ex
R1(config)#ip ips notify log
R1(config)#ex
R1#
 %IPS-6-ENGINE_BUILDS_STARTED:  00:14:37 UTC Mar 01 1993

 %IPS-6-ENGINE_BUILDING: atomic-ip - 3 signatures - 1 of 13 engines

 %IPS-6-ENGINE_READY: atomic-ip - build time 8 ms - packets for this engine will be scanned

 %IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 8 ms

%SYS-5-CONFIG_I: Configured from console by console
```

7. Click on server → services → NTP → change time.



8. Again click on R1 → CLI → Type Command.
   - clock set 02:56:02 31 jan 2026
   - conf t
   - service timestamps log datetime msec
   - loggin host 10.10.10.2
   - ip ips signature-definition
   - signature 2004 0
   - status
   - retired true
   - enabled true
   - retired false
   - ex
   - engine
   - event-action produce-alert
   - event-action deny-packet-inline
   - ex
   - ex
   - ex
     - When asked for confirm hit enter

```
R1#clock set 02:56:02 31 jan 2026
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#service timestamps log datetime msec
R1(config)#loggin host 10.10.10.2
R1(config)#ip ips signature-definition
R1(config-sigdef)#signature 2004 0
R1(config-sigdef-sig)#status
R1(config-sigdef-sig-status)#retired true
R1(config-sigdef-sig-status)#enabled true
R1(config-sigdef-sig-status)#retired false
R1(config-sigdef-sig-status)#ex
R1(config-sigdef-sig)#engine
R1(config-sigdef-sig-engine)#event-action produce-alert
R1(config-sigdef-sig-engine)#event-action deny-packet-inline
R1(config-sigdef-sig-engine)#ex
R1(config-sigdef-sig)#ex
R1(config-sigdef)#ex
Do you want to accept these changes? [confirm]
%IPS-6-ENGINE_BUILDS_STARTED:
%IPS-6-ENGINE_BUILDING: atomic-ip - 303 signatures - 3 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 480 ms - packets for this engine will be scanned
%IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 648 ms

R1(config)#
```

Copy Paste

Top

9. Click on PC → Desktop → command prompt → write command.

- ping 10.10.10.2

PC0

Physical Config Desktop Programming Attributes

Command Prompt X

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

10. Click on server → services → SYSLOG

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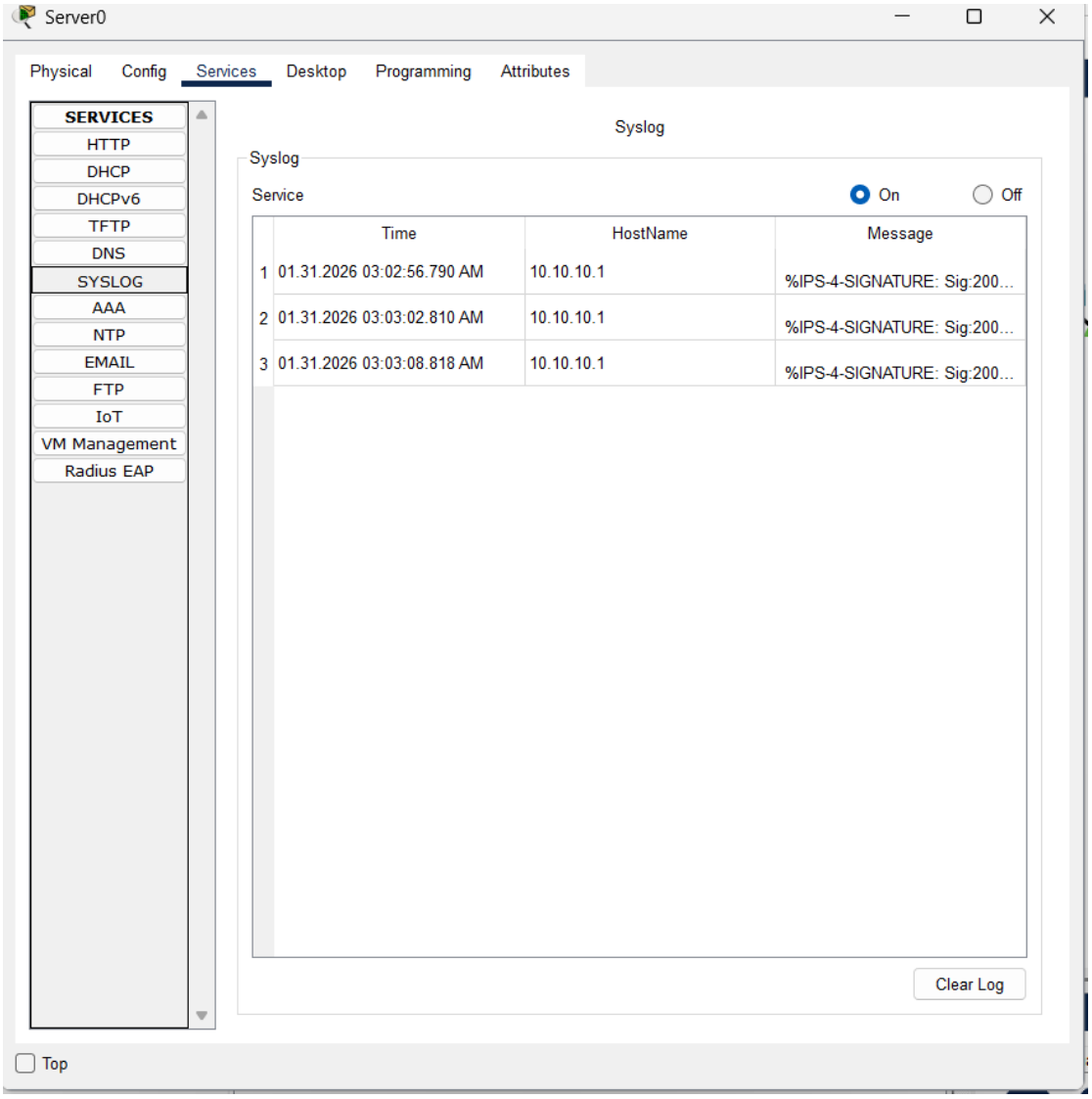

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